GENOXA

Interactive Bulk RNA-Seq Analysis Platform

Version 0.1

USER MANUAL

A no-code guide for uploading your own RNA-Seq count data, differential expression, gene ontology and KEGG over-representation, and machine-learning based gene modeling

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1. Overview

GENOXA is an interactive, no-code web application for analyzing bulk RNA-Seq gene expression data. It combines traditional bioinformatics workflows (limma and edgeR) with machine-learning based gene modeling, so that users without programming experience can go from their own read-count files to a finished, shareable analysis report.

In Version 0.1 you supply the data yourself: you upload a read counts file and a sample information file (and, optionally, a clinical/annotation file) from the **Upload Files** section of the left sidebar.

1.1 Who this manual is for

This manual is written for researchers, students, and analysts who want to run a differential expression and/or machine-learning analysis on their own bulk RNA-Seq count data without writing R code. Familiarity with basic RNA-Seq concepts (counts, normalization, fold-change, p-values) is helpful but not required — each parameter is explained in plain language in this manual and inside the app's own "User Manual" tab.

1.2 Key features

- Traditional bioinformatics results for bulk RNA-Seq data using limma and edgeR.
- Interactive, publication-quality visualizations, downloadable as .png or .pdf with customizable size and resolution.
- Multiple color palettes and styling options (titles, axis labels, legends) for plots.
- Comprehensive Gene Ontology (GO) over-representation analysis, including bar, dot, gene-concept network, heat, enrichment map, and UpSet plots, plus KEGG pathway enrichment and Pathview diagrams.
- Machine-learning based gene modeling with performance, variable-importance, and ROC plots.
- One-click, parameterized analysis report, downloadable as HTML, PDF, or Word.

1.3 GENOXA workflow at a glance

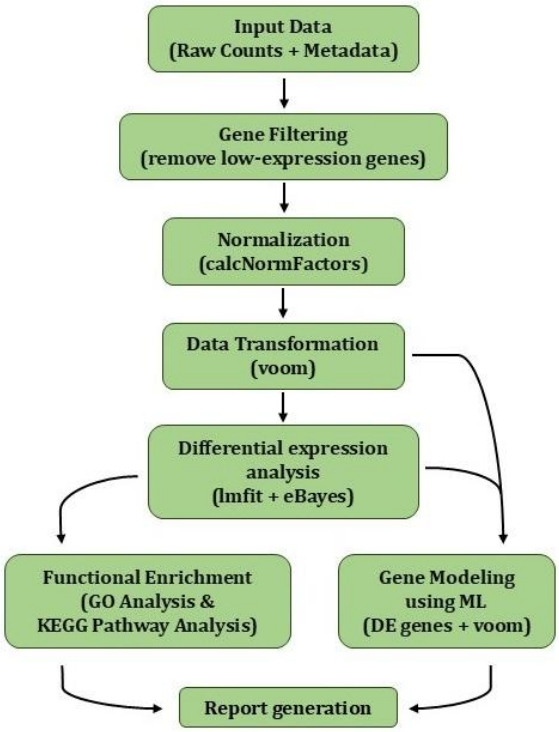


Figure 1. GENOXA workflow (also shown on the app's About page).

GENOXA follows a fixed pipeline:

1. **Upload Files** — upload a read counts file, a sample information file and, optionally, a clinical/annotation file.
2. **Data Preprocessing** — filter low-count genes and normalize the count matrix.
3. **Differential Expression (DE) Analysis** — apply the voom transformation, fit a linear model (lmFit + eBayes), and identify significantly up- and down-regulated genes between two groups.
4. **Gene Modeling using Machine Learning (ML)** — build and compare predictive models using the normalized expression of the DE genes as features.
5. **GO and KEGG Over-Representation Analysis (ORA)** — test whether the top DE genes are enriched for specific GO terms and KEGG pathways.
6. **Results** — browse and download every table and plot produced by the steps above.
7. **Report** — generate a single document summarizing the methods, parameters, and key results.

> ***Note:*** *Clicking **Start Analysis** at the bottom of the sidebar runs steps 2–4 in one pass, using the parameters currently set in the sidebar. GO and KEGG analysis (step 5) runs separately, when you click **Update Results** under Results › ORA results › GO over-representation (see Section 4.2).*

1.4 Finding your way around the app

Everything is reached from the left sidebar:

Sidebar item	What it contains
About	Welcome page: overview, key features, the list of results provided, the workflow diagram, developer contact details, and news.
User Manual	In-app help with five tabs: Upload Files, Data Preprocessing, Differential expression analysis, GO over-representation, and Gene modeling. The Upload Files tab has links to download the demo files.
Upload Files	The three file upload boxes and the clinical/annotation Yes/No choice (Section 2).
Data Preprocessing, Differential expression analysis, GO over-representation, Gene Modeling using ML	Parameters for each analysis stage (Section 3).
Results	All tables and plots, plus the Report tab (Sections 5–7).
Start Analysis	The button that runs the analysis (Section 4).

2. Getting Started: Preparing and Uploading Your Data

GENOXA Version 0.1 analyzes data that you upload. Prepare your files in the layout described below before you start — files that don't follow it are the most common cause of empty tables or error messages.

2.1 Opening GENOXA

GENOXA is an R Shiny application that runs in a web browser. If you were given a link to a running copy, open it in your browser; the About page appears first.

To run GENOXA on your own computer, install R together with the CRAN and Bioconductor packages loaded in **01_library.R** (including the organism annotation packages, such as org.Hs.eg.db), start R in the GENOXA folder, and run **shiny::runApp()**. The app opens in your default browser.

2.2 Input files

File	Required?	Contents and layout
Read counts file	Yes	Raw (un-normalized) read counts, with one row per gene and one column per sample. The first column holds the gene identifiers and the top row holds the sample names.
Sample information file	Yes	Two columns: the sample names in the first column and the group (condition) of each sample in the second — for example, normal vs. cancer tissue or treated vs. untreated. Include a header row.
Clinical/Annotation data file	Optional	The sample names in the first column, followed by one column per additional variable, such as race, age, cell type, color, length, or shape. Used to add annotation tracks to the DE heat map.

2.3 File format rules

- **Tab-delimited text.** Save each file as tab-delimited text (.txt is recommended). The upload boxes also accept .csv, but GENOXA reads every file with tabs as the separator, so a comma-separated .csv will not be read correctly.
- **Header row.** Each file needs a header row. The top-left cell, above the gene IDs or sample names, may be left blank (as in the demo files).
- **Same sample order in every file.** GENOXA matches samples by position, not by name: row 1 of the sample information file (and of the clinical/annotation file) must describe column 1 of the counts file, row 2 must describe column 2, and so on.
- **Two groups only.** Version 0.1 supports two-group comparisons (e.g., treated vs. untreated or normal vs. cancer tissue).
- **One column per sample.** GENOXA does not support biological and/or technical replicate read counts; format your data so that each column is a single sample.
- **Gene identifiers.** Use one identifier type throughout (for example, Ensembl gene IDs or gene symbols) and select the matching Key Type for GO analysis (Section 3.3). Version suffixes on IDs (such as the ".15" in ENSG00000000003.15) are removed automatically before GO and KEGG analysis.

- **Enough samples per group.** For the ML step, each group must be represented in both the training and test sets; otherwise the ROC plot and confusion matrix cannot be produced.

2.4 Example file layouts

The first rows of the demo files show the expected layout (the top-left header cell is blank).

Read counts file

	Sample_1	Sample_2	Sample_3	…
ENSG00000000003.15	5405	3411	2628	…
ENSG00000000005.6	11	55	6	…
ENSG00000000419.13	1699	1158	1010	…

Sample information file

	Group_Type
Sample_1	Primary solid Tumor
Sample_2	Solid Tissue Normal
Sample_3	Primary solid Tumor

Clinical/annotation data file

	race	Group_Type	age
Sample_1	white	Primary solid Tumor	53
Sample_2	white	Solid Tissue Normal	53
Sample_3	asian	Primary solid Tumor	66

2.5 Demo data

Three demo files can be downloaded from **User Manual › Upload Files** inside the app. Use them to try GENOXA or as templates for your own files:

- **Demo_counts.txt** — raw read counts for 60,660 genes (Ensembl gene IDs) across 10 samples.
- **Demo_metadata.txt** — the group of each sample (Primary solid Tumor or Solid Tissue Normal).
- **Demo_clinical_or_annotation.txt** — race, group, and age for each sample.

2.6 Step-by-step: uploading your files

1. In the left sidebar, click **Upload Files** to expand it.
2. Under **1. Upload a read counts file**, click Browse and choose your counts file.
3. Under **2. Upload a sample information file**, choose your sample information file. The **Select reference group for DE** list (in the Differential expression analysis section) is filled with the groups from this file.

4. For **Do you want to add Clinical/Annotation data ??**, choose **No** to skip, or **Yes** to show **3. Upload your Clinical/Annotation data file** and upload it.
5. Open each parameter section in the sidebar and review its settings (Section 3).
6. Click **Start Analysis** at the bottom of the sidebar (Section 4).

> ***Note:*** *Choose Yes for clinical/annotation data before you click Start Analysis. The Clinical/Annotation data table and the annotated heat map are only available when Yes was selected and a file was uploaded.*

3. Setting Analysis Parameters (Left Sidebar)

After uploading your files, use the remaining sidebar sections to configure each analysis step; each section expands when you click it. Every parameter has a more detailed explanation inside the app, under the User Manual tab. The defaults listed below are the values GENOXA starts with.

3.1 Data Preprocessing

Parameter	What it does
Filter raw reads by rowSum	Keeps genes whose total read count across all samples is at least this value; genes below it are removed before any other step (default 5).
Filter raw reads by FilterByExpr	The minimum count (min.count) used by edgeR's filterByExpr(), which removes genes that are not expressed at a worthwhile level in enough samples, improving the robustness of DE testing (default 5).
Select Normalization method for calcNormFactors	How library-size and composition differences between samples are corrected: TMM (default), RLE, upperquartile, or none.

Normalization method choices

- **TMM (Trimmed Mean of M-values):** corrects for library-size and compositional differences; best default choice for most RNA-Seq datasets.
- **RLE (Relative Log Expression):** scales samples to a common median relative log expression; assumes most genes are not differentially expressed.
- **Upper Quartile:** scales by the upper quartile of counts; less sensitive to a few highly-expressed genes.
- **None:** no normalization; use only if the data was already normalized elsewhere.

3.2 Differential Expression (DE) Analysis

Parameter	What it does
Select reference group for DE	The baseline/control group (e.g., Normal) that the other group is compared against, set with relevel(). The list is filled from your sample information file, and the first group listed is selected by default. A positive log2FC means higher expression in the other group than in the reference group.
Cutoff value for p-values and adjusted p-values	Significance threshold for DE genes (default 0.05). It is applied to the p-values after the selected adjustment method; choose none as the adjustment method to apply it to raw p-values.
Minimum absolute log2-fold change (log2FC) to use as threshold for differential expression	Genes must change by more than this amount, in either direction, to be called differentially expressed (default 1, i.e., log2FC > 1 or < −1).
Select adjustment method	Multiple-testing correction applied to p-values: none, bonferroni, holm, BH (default), fdr, or BY.

> ***Note:*** *If you change the DE cutoff, log2FC threshold, or adjustment method after the analysis has run, the DE table, volcano plot, and heat map update straight away, but the ML results do not. Click Start Analysis again so the models use the new DE gene list.*

3.3 GO Over-Representation (ORA)

Parameter	What it does
Select Organism Database	Gene annotation database for your species. 17 organisms are available (listed below); the default is Human (org.Hs.eg.db). The same organism is used for KEGG.
Select Key Type	Identifier type of the gene names in your counts file: ACCNUM, ALIAS, ENSEMBL (default), ENSEMBLPROT, ENSEMBLTRANS, ENTREZID, GENENAME, REFSEQ, SYMBOL, UNIGENE, or UNIPROT. Once the analysis starts, the list is updated to the key types available in the selected organism database.
Select subontology	BP (Biological Process), MF (Molecular Function), CC (Cellular Component), or ALL (default).
Minimum / Maximum Gene Set Size	Only GO terms whose gene sets contain between these numbers of genes are tested (defaults 10 / 500).
P-Value Cutoff / q-Value Cutoff	Significance thresholds for enriched GO terms (defaults 0.05 / 0.02).
Adjust Method	Multiple-testing correction for GO term p-values: none, holm, hochberg, hommel, bonferroni, BH (default), BY, or fdr.

KEGG enrichment uses the same organism, p-value and q-value cutoffs, and minimum/maximum gene set sizes.

Supported organisms

Organism	Database	Organism	Database
Human	org.Hs.eg.db	Worm	org.Ce.eg.db
Mouse	org.Mm.eg.db	Chicken	org.Gg.eg.db
Rat	org.Rn.eg.db	Canine	org.Cf.eg.db
Yeast	org.Sc.sgd.db	Pig	org.Ss.eg.db
Fly	org.Dm.eg.db	E. coli strain K12	org.EcK12.eg.db
Arabidopsis	org.At.tair.db	Xenopus	org.Xl.eg.db
Zebrafish	org.Dr.eg.db	Chimp	org.Pt.eg.db
Bovine	org.Bt.eg.db	Anopheles	org.Ag.eg.db
		E. coli strain Sakai	org.EcSakai.eg.db

3.4 Gene Modeling using Machine Learning

Parameter	What it does
Enter The size of the training data set	Proportion of samples used to train the models; the remaining samples form the test set used for the ROC plot and confusion matrix (default 0.80, i.e., 80% train / 20% test). The split keeps the group proportions similar in both sets.
Seed	Random seed for the train/test split, ensuring reproducible results (default 8314).
Select resampling method to use	How the training data is resampled to validate the models during training: none, boot, boot632, optimism_boot, boot_all, cv, repeatedcv (default), LOOCV, LGOCV, oob, timeslice, adaptive_cv, adaptive_boot, or adaptive_LGOCV.
The number of complete sets of folds to compute	Shown only for repeatedcv: how many times the whole cross-validation is repeated (default 3).
Number of resampling iterations	The number of folds (for cv and repeatedcv) or resampling iterations (for the bootstrap methods) (default 10).
Random number seed	Seed used during model training, for reproducibility (default 351).
Select ML algorithms	The algorithms to train and compare (see 3.5). Select a minimum of two and a maximum of four. Selected by default: Naive Bayes, Flexible Discriminant Analysis, Support Vector Machines, and Boosted Generalized Linear Model.

> ***Note:*** *GENOXA enforces the two-to-four algorithm rule: selecting fewer than two or more than four models shows the message "ML model options! Please select a minimum of two ML models." (or "…a maximum of four ML models."), and the analysis will not start until the selection is corrected.*

3.5 Available ML algorithms

Algorithm	Description
Naive Bayes (naive_bayes)	Probabilistic classifier assuming conditional independence between features; fast and effective for high-dimensional data.
Flexible Discriminant Analysis (fda)	Extension of Linear Discriminant Analysis using non-linear transformations for complex class boundaries.
Support Vector Machines (svmRadial)	Finds an optimal separating boundary using a radial basis function kernel; models complex non-linear relationships.
Boosted Generalized Linear Model (glmboost)	Boosting-based Generalized Linear Model that combines weak learners to reduce overfitting.
Random Forest (rf)	Ensemble of decision trees; robust to overfitting and provides variable-importance estimates.
C5.0	Advanced, interpretable decision-tree algorithm supporting boosting and rule-based models.
glmnet	Regularized GLM using Lasso (L1), Ridge (L2), or Elastic Net penalties for feature selection on high-dimensional data.
Partial Least Squares (kernelpls)	Dimensionality-reduction regression technique suited to highly correlated predictors.

All models are trained with the caret R package. For details on each algorithm, see the caret model list: topepo.github.io/caret/available-models.html.

Important Disclaimer: machine-learning results are generated by automated statistical models and should be validated using additional independent methods before any conclusions are drawn. The developers do not take responsibility for claims or decisions made solely on the basis of ML results produced by this application.

4. Running the Analysis

4.1 Start Analysis

After uploading your files and reviewing every parameter in the sidebar, click **Start Analysis** at the bottom of the sidebar. GENOXA will:

- Switch the view to the Results tab.
- Show the message "Please wait patiently... Your Data analysis has started..." — wait until the tables and plots appear on screen.
- Run filtering, normalization, the voom transformation, differential expression testing, and ML gene modeling in sequence, using all available processor cores for parallel processing.

Large datasets, many resampling iterations, and slower algorithms (such as Random Forest or Support Vector Machines) increase the run time.

> ***Note:*** *Version 0.1 does not check your files before running. If a table or plot stays empty or shows an error message, check your files against Section 2.3 and see Section 8.*

4.2 Running GO and KEGG analysis

GO and KEGG over-representation analysis runs on demand, after Start Analysis has finished:

1. Go to **Results › ORA results › GO over-representation**.
2. In **Number of Genes to Select for GO analysis**, enter how many of the top DE genes (ranked by p-value) to analyze. The default is 10; the total number of DE genes found is shown below the box.
3. Click **Update Results**. The message "The GO analysis is in progress. Please wait..." appears.
4. When the GO table appears, open the other ORA sub-tabs to view the GO plots, the KEGG results, and the KEGG Plot.

> ***Note:*** *GO and KEGG settings are read from the sidebar each time you click Update Results, so you can change them (or the number of genes) and click Update Results again without re-running the whole analysis. KEGG enrichment and Pathview diagrams need an internet connection to reach the KEGG database.*

5. Results Provided

The Results tab is organized into seven sub-tabs: **User Data Information**, **QC and Data Normalization**, **Exploratory data analysis**, **DE results**, **Gene modeling**, **ORA results**, and **Report** (Section 7). Most plots have on-screen customization controls (plot type, color palette, titles, axis labels, legend) and a separate set of options — height, width, resolution (dpi), and file type (png or pdf) — that apply only to the downloaded copy of the plot.

Plots that show samples or genes have First and Last boxes that set the range plotted (1 to 10 by default). To plot all samples or genes, replace 10 with the total shown below the boxes.

5.1 User Data Information

Output	Description
Raw Read Counts Data	Interactive table of the uploaded counts, with the total number of rows (genes) and columns (samples).
Sample and Group Information	Table of sample names and their groups, with row and column totals.
Raw Read Counts Plot	Boxplot, column, or violin plot of counts per sample, colored by group. Choose Count or Log2 Count (log2 of count + 1; default), the sample range, the color palette (viridis, magma, inferno, plasma, cividis, rocket, mako, or turbo), and whether to show the title, axis labels, and legend.
Clinical/Annotation data	Table of the uploaded clinical/annotation file, with row and column totals. Shown only when you chose Yes for clinical/annotation data.

5.2 QC and Data Normalization

Output	Description
Filtered Read Counts Plot	Counts per sample after the rowSum filter, with the number of genes retained. Same plot type, transformation, and styling options as the Raw Read Counts Plot.
FilterByExpr Plot	Counts per sample after edgeR's filterByExpr() filter, with the number of genes retained.
Before and after filtering data	Side-by-side density curves of log-CPM values for each sample: (A) raw counts and (B) counts after FilterByExpr, with the number of genes in each. Titles and axis labels can be set separately for each panel.
Data Normalization Plot	Distribution of the normalized, voom-transformed expression values (log2-CPM) per sample after calcNormFactors() normalization, as a boxplot, column, or violin plot.

5.3 Exploratory Data Analysis

Output	Description
PCA Plot	Principal Component Analysis scatter plot (PC1 vs. PC2) of the normalized expression values, colored by group, to reveal overall sample structure.

Output	Description
MDS Plot	Multi-Dimensional Scaling plot (leading log-fold-change dimensions 1 and 2), colored by group.
Voom Plot	limma-voom mean-variance trend plot, assessing whether the data is ready for linear modeling. Download options only.
Mean-variance trend Plot	Final-model mean-variance relationship after model fitting (plotSA). Download options only.

5.4 DE Results

Output	Description
DE table	The significant DE genes only — those passing the log2FC threshold and the p-value cutoff — sorted by p-value, with logFC, AveExpr, t, P.Value, adj.P.Val, and B. An info box reports the total number of significant DE genes and how many are up- and down-regulated. Five download buttons are described in Section 6.2.
Volcano Plot	log2 fold change vs. significance (−log10 adjusted p-value by default, or −log10 p-value). Options: dotted lines marking the log2FC threshold and p-value cutoff, labels for the top hits (default 10, text size 15), the color of significant genes (green, red, gray, teal, purple, orange, blue, or magenta), and the plot title. Non-significant genes are shown in brown.
DE Heat Map Plot	Clustered heat map of the normalized expression of selected DE genes (in DE-table order) across selected samples. Options: gene and sample range, title, font size, color palette (RdYlBu, BluYl, Greens, Oranges, Reds, YlOrBr, YlOrRd, BuPu, GnBu, or OrRd), scaling by row (default), column, or none, border color, row/column clustering, legend, and row/column names. With clinical/annotation data, Show Clinical/Annotation data adds a color track for each clinical variable.

5.5 Gene Modeling (ML) Results

Models are trained on the normalized (voom) expression of all significant DE genes to predict each sample's group. Group names are converted to R-compatible names for modeling — spaces become dots, so "Primary solid Tumor" appears as "Primary.solid.Tumor" in the ML outputs. An algorithm that fails on your data is skipped and does not appear in the results.

Output	Description
Performance Plot	Mean ± standard error of Accuracy and Kappa across the resampling iterations for each trained model. Options: title, color palette, and text size. Download Performance matrix saves the best Accuracy and Kappa for each model, sorted by Accuracy (CSV).
Variable Importance Plot	The most important genes for each model, as a lollipop (default) or bar chart. Options: number of top genes (default 10), rows and columns of the plot grid (default 2 × 2), title, text size, and palette.

Output	Description
ROC Plot	Receiver Operating Characteristic curves for each model on the held-out test set. With Multiple Plots checked (default), each model gets its own panel showing its AUC (%); uncheck it to overlay all curves in one plot. Options: title, text size, and palette. Download Confusion matrix saves the test-set predictions vs. the true groups for each model (CSV).

5.6 ORA Results (Gene Ontology / KEGG)

These sub-tabs fill in after you click Update Results (Section 4.2).

Output	Description
GO over-representation	enrichGO() results table for the top N DE genes (N is set in Number of Genes to Select for GO analysis; default 10), downloadable as CSV.
Bar Plot	Enriched GO terms as bars (gene count, colored by adjusted p-value). Options: number of GO categories shown (default 10), font size (default 8), and title.
Dot Plot	Enriched GO terms as dots (gene ratio, sized by gene count); split into BP, MF, and CC panels when Subontology = ALL. Options: GO categories (default 10), font size (default 8), and title.
Gene-Concept Network Plot	Network linking enriched GO terms to their associated genes (cnetplot), after redundant terms are removed with simplify(). Options: GO categories (default 5), coloring edges by enriched terms, circular layout (on by default), and title.
Heat Plot	Gene-by-GO-term heat map of enrichment results. Options: GO categories (default 5) and title.
Enrichment Map Plot	Network of enriched GO terms connected by shared-gene similarity (emapplot). Options: GO categories (default 30) and title.
UpSet Plot	Set-intersection plot showing gene overlap across enriched GO terms (after simplify()). Options: GO categories (default 10) and title.
KEGG Results	enrichKEGG() results for the same top N genes after conversion to Entrez IDs, downloadable as CSV.
KEGG Plot	Pathview diagram for a pathway chosen from Select KEGG Pathway ID (filled from the KEGG results). Click Generate Pathview; once it is drawn, Download .png (KEGG-style diagram) and Download .pdf (graph-layout version) buttons appear.

6. Downloading Plots, Tables, and Data

6.1 Plots

Nearly every plot in GENOXA has its own **Download Plot** button, alongside independent controls for:

- Height and width of the exported image, in inches (defaults 8 and 12).
- Resolution in dpi (default 300; applies to png files).
- File type — png (default) or pdf.

These download settings apply only to the exported file; they do not change how the plot appears on screen. Files are named after the plot, for example Volcano_plot.png or GO_dot_plot.pdf. The KEGG Plot is the exception: it has its own Download .png and Download .pdf buttons and no size options.

6.2 Tables

Data tables are downloaded as CSV files from buttons beneath each table:

Button	Where	File and contents
Download all results	DE results › DE table	Differential_expression_analysis_all_data.csv — all significant DE genes.
Download Top Hits results	DE results › DE table	Top_Differential_expression_analysis.csv — the top N genes, set in the Top hits box (default 10).
Download Up and Down regulated genes results	DE results › DE table	Up_and_Down_regulated_genes.csv — significant genes with an added Gene_Expression column (Up-regulated or Down-regulated).
Download Up-regulated genes only	DE results › DE table	Up_regulated_genes.csv
Download Down-regulated genes only	DE results › DE table	Down_regulated_genes.csv
Download Performance matrix	Gene modeling › Performance Plot	Model_performance_results.csv — Accuracy and Kappa for each model.
Download Confusion matrix	Gene modeling › ROC Plot	ML_Confusion_matrix.csv — Prediction, Reference, Value, and Method for each model.
Download GO over-representation results	ORA results › GO over-representation	GO_over_representation_results.csv
Download KEGG results	ORA results › KEGG Results	KEGG_results.csv

7. Generating the Analysis Report

The **Report** sub-tab (the last tab under Results) produces a single document, titled "GENOXA Ver 0.1 Report", that documents your analysis. It contains:

- **Method details** — the packages used (edgeR, limma, clusterProfiler, and caret) and a methods description filled in with your parameter values.
- **Results** — the number of genes and samples analyzed, the genes retained after each filtering step, and the numbers of significant, up-regulated, and down-regulated DE genes.
- **Quick summary of parameters** — every Data Preprocessing, DE, GO, and ML setting.
- **References.**

The report does not include the plots or tables themselves; download those separately (Section 6).

7.1 How to generate a report

1. Run the analysis (Section 4).
2. Go to **Results › Report**.
3. Under **Select Report Format**, choose PDF (default), Word, or HTML.
4. Click **Download Data analysis Report**. The file is saved as GENOXA_report_<date>.pdf, .docx, or .html.

> ***Note:*** *The report reads the parameter values currently set in the sidebar when you click download. If you change a parameter after the analysis, click Start Analysis again before downloading so the report matches the results on screen. PDF reports need a LaTeX installation (such as TinyTeX) on the computer running GENOXA; if the PDF fails, choose Word or HTML.*

8. Troubleshooting & Common Messages

Message / Symptom	Likely cause and fix
Select reference group for DE list is empty	The sample information file has not been uploaded yet, or its extension is not .txt or .csv. Upload it under Upload Files; the list fills with the groups in its second column.
Tables or plots stay empty, or show an error, after Start Analysis	A file is not tab-delimited, has no header row, or the counts file contains non-numeric values. Re-save the files as tab-delimited text (Section 2.3) and run the analysis again.
"undefined columns selected" or a similar error under a plot	The Last sample or Last DE gene value is larger than the number available. Set it to no more than the total shown below the boxes.
Samples appear in the wrong group, or DE results look unexpected	The rows of the sample information file are not in the same order as the columns of the counts file. GENOXA matches samples by position, not by name — reorder the file and re-run.
"ML model options! Please select a minimum of two ML models." / "…a maximum of four ML models."	Fewer than two or more than four ML algorithms are selected. Adjust the selection in the Gene Modeling using ML section of the sidebar.
"Total 0 significantly differentially expressed genes were identified."	No genes passed the current p-value cutoff and log2FC threshold. Loosen them in the sidebar and click Start Analysis again; the heat map, ML models, and GO/KEGG analysis all depend on DE genes.
An ML model is missing from the Performance, Variable Importance, or ROC plot	That algorithm could not be trained on your data (or importance could not be computed for it) and was skipped. Choose a different algorithm, or add samples.
ROC plot or confusion matrix is not produced	The test set is missing samples from one group, usually because a group is very small. Add samples, or adjust the training data set size.
GO table or GO plots are empty	Update Results has not been clicked; the Key Type or organism does not match your gene IDs; too few genes were selected; or the cutoffs are too strict (the default q-value cutoff is 0.02). Check the Key Type, increase the number of genes, or relax the cutoffs, then click Update Results again.
Select KEGG Pathway ID list is empty	No KEGG pathway was enriched, the gene IDs could not be converted to Entrez IDs, or there is no internet connection to KEGG.
"Error generating Pathview plot: …" or "PNG file not generated."	Pathview could not download or draw the selected pathway. Check the internet connection, or choose another pathway ID.
Heat map with clinical/annotation data fails	The clinical/annotation file must have exactly one row per sample, in the same order as the columns of the counts file. Uncheck Show Clinical/Annotation data to confirm the heat map works without it.
Report download fails or the file is empty	A PDF report needs LaTeX (for example, TinyTeX) on the computer running GENOXA. Choose Word or HTML instead, and make sure the analysis has finished first.

9. About GENOXA

GENOXA is developed and maintained by Sagar Patel, Ph.D.

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9.1 News and updates

- February 26, 2026 — GENOXA Ver 0.2 selected as a winning tool in the National Cancer Institute (NCI) Genomic Data Commons (GDC) Analysis Tool Challenge (2025), recognizing its innovation in integrating advanced analytical tools with GDC infrastructure.
- May 8, 2025 — GENOXA Ver 0.2 released, featuring integration with TCGA datasets from the GDC portal as part of the GDC Tool Challenge submission.
- November 25, 2024 — GENOXA Ver 0.1 released.